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Bill of Materials — Solar Rooftop 108.55 kWp

A hotel in Roi Et · reviewed from the construction drawing set (For Construction, Aug 2026) · Rev03, 31 October 2026

PV modules	Inverters	Optimizers	Mounting
167 × 650 Wp	**2 × 50 kW**	**84**	**72 rails · 406 L-feet**
LONGi LR7-72HVH-650M	Huawei, 100 kWac	1 per 2 modules	metal-sheet roof

Drawing read from the drawings Est. Solar ESS estimate Not given not on the drawings, measure on site. Mounting quantities are counted from the EE-05 layout: south roof 18 landscape columns (6 × 4, 12 × 5 modules, rails up-slope); north roof 16 portrait rows (12 × 5, 2 × 4, 2 × 2, rails along the eave) plus east-face columns of 8 and 3; 2 rails per run; module 2,382 × 1,134 mm with 20 mm gaps.

1 PV modules & inverters

No.	Item	Specification	Qty	Unit	Basis	Sheet
1	**PV module**	LONGi LR7-72HVH-650M, 650 Wp (Vmp 44.56 V, Imp 14.59 A, Voc 53.9 V, Isc 15.29 A)	**167**	pcs	Drawing	GE-03, EE-01–EE-06
2	**Module optimizer**	1 per 2 modules (OPT1…OPT11 per string); model not specified	**84**	pcs	Drawing	GE-03, EE-07, EE-08
3	**Grid-tie inverter, 3-phase**	Huawei 50 kW, 400 V 3W+N+PE, max 84 A (model written SUN2000-50KTL-MC0 / -50K-MC0 / -50K-MPO)	**2**	pcs	Drawing	EE-01–EE-04, EE-12

2 DC side

No.	Item	Specification	Qty	Unit	Basis	Sheet
4	**PV string cable**	Solar cable 1C-6 mm², red/black pair per string, roof to inverter room	**—**	m	Not given	EE-02, EE-03
5	**String earth conductor**	1C-16 mm² IEC01, run with the strings	**—**	m	Not given	EE-02, EE-03
6	**DC string fuse**	DC fuse 1,500 V 35 A, one per string	**8**	pcs	Drawing	EE-02, EE-03
7	**DC fuse box**	Box Fuse No.1 / No.2 in inverter room; enclosure size not given	**2**	pcs	Drawing	EE-12
8	**MC4 connector pair**	At string ends (plus spares)	**16**	pairs	Est.	EE-07, EE-08

3 AC panel (SMDB)

No.	Item	Specification	Qty	Unit	Basis	Sheet
9	**Main breaker**	MCCB 3P 250 AT / 250 AF; kA not given	**1**	pcs	Drawing	EE-01
10	**Inverter feeder breaker**	MCCB 3P 100 AT / 100 AF; kA not given	**2**	pcs	Drawing	EE-01
11	**Spare way**	MCCB 3P space (F3)	**1**	way	Drawing	EE-01
12	**Busbar set**	Cu busbar 3-phase, full N 100%, G 50%; size not given	**1**	set	Drawing	EE-01
13	**Surge protection**	Type 2, 50 kA, with 3P fuse	**1**	set	Drawing	EE-01
14	**Panel meter**	Digital panel meter (PDM)	**1**	pcs	Drawing	EE-01
15	**Current transformer**	CT 250/5 A	**3**	pcs	Drawing	EE-01
16	**Control fuse**	Fuse 4P 6 A for lamps and meter	**1**	pcs	Drawing	EE-01
17	**Pilot lamp**	White	**3**	pcs	Drawing	EE-01
18	**Emergency stop + reset push buttons**	Ratings not given	**1**	set	Drawing	EE-01
19	**SMDB enclosure**	About 800 W × 1,600 H mm; depth and IP not given	**1**	pcs	Drawing	EE-12
20	**Inverter → SMDB cable**	4 × 1C-25 mm² 0.6/1 kV CV-FD + G 16 mm² THW (EE-01 says 16 mm² + G 10)	**—**	m	Not given	EE-01–EE-03

21	**SMDB → existing MDB feeder**	2 × (4 × 1C-25 mm²) 0.6/1 kV CV-FD + G 16 mm² THW	—	m	Not given	EE-01
22	**Export-control power meter**	UMG96RM at existing MDB, with aux and voltage fuses	**1**	set	Drawing	EE-01, EE-04
23	**CTs at MDB for meter**	Ratio not given	**3**	pcs	Est.	EE-01
24	**Meter CT cable**	3 × (CVVS 2C-2.5 mm²)	—	m	Not given	EE-01
25	**Meter voltage cable**	VCT 3C-2.5 mm²	—	m	Not given	EE-01

4 Roof mounting

No.	Item	Specification	Qty	Unit	Basis	Sheet
26	**Aluminium rail**	Solar mounting rail, 6.0 m bars; 72 rails: 48 × 5.75 m, 16 × 4.6 m, 4 × 2.29 m, 2 × 9.2 m, 2 × 3.44 m (≈ 384 m + 5% spare)	**72**	bars	Est.	EE-05, EE-11
27	**Rail splice + bolts**	Only the two 9.2 m runs need a joint with 6.0 m bars (≈ 68 if 4.4 m bars are used)	**2**	sets	Est.	EE-05, EE-11
28	**Mid clamp**	Aluminium, M8 bolt + T-nut; 2 × (167 modules − 36 rail runs)	**262**	pcs	Est.	EE-05, EE-11, EE-14
29	**End clamp**	Aluminium, M8 bolt + T-nut; 4 per rail run × 36 runs	**144**	pcs	Est.	EE-05, EE-11
30	**Grounding plate**	Under each mid clamp	**262**	pcs	Est.	EE-14
31	**Rail earth lug**	1 per rail, bonded with 1C-16 mm² IEC01	**72**	pcs	Est.	EE-14
32	**L-foot with EPDM rubber**	Aluminium, ≈ 1 per 1.2 m of rail (confirm against roof purlin spacing)	**406**	pcs	Est.	EE-05, EE-11
33	**Self-drilling screw with O-ring**	2 per L-foot + 5% spare	**853**	pcs	Est.	EE-11
34	**Rail end cap**	2 per rail	**144**	pcs	Est.	EE-11
35	**Cable clip**	For module and optimizer leads, ≈ 2 per module	**340**	pcs	Est.	EE-07
36	**Walkway grating**	HDG expanded-metal grating W300 × L2400, frame L38×38×3, H 40 mm	**13**	pcs	Est.	EE-10, EE-11
37	**DC cable tray**	Beside walkway; size and length not given	—	m	Not given	EE-11, EE-14
38	**IMC conduit (roof)**	Size not given; roof run only	**32**	m	Est.	EE-09
39	**PPR water pipe (cleaning) + U-bolts HDG**	Size, length and fittings not given	—	m	Not given	EE-11

5 Monitoring

No.	Item	Specification	Qty	Unit	Basis	Sheet
40	**Data logger**	Huawei SmartLogger 5000 with power adapter	**1**	set	Drawing	EE-01, EE-04
41	**Logger enclosure**	About 600 mm wide	**1**	pcs	Drawing	EE-12
42	**RS485 cable**	Inverters + meter to logger; type not given	—	m	Not given	EE-01, EE-04
43	**LAN cable**	CAT5 (EE-01) / FTP CAT6 (EE-04)	—	m	Not given	EE-01, EE-04

6 Earthing

No.	Item	Specification	Qty	Unit	Basis	Sheet
44	**Ground rod**	Copper-bonded Ø 5/8" × 3.00 m	**3**	pcs	Drawing	EE-13, EE-15
45	**Ground inspection pit + test box**	Test box 25 cm above floor	**1**	set	Drawing	EE-15

46	**Earth conductor 25 mm²**	1 × 25 mm² IEC01 (G): rods → MDB → inverter station	—	m	Not given	EE-13
47	**Earth conductor 16 mm²**	1 × 16 mm² IEC01 (G): inverters, rack, roof; in PVC Ø 3/4" with saddle clamps	—	m	Not given	EE-13, EE-15
48	**Tray bonding jumper**	Cable lug + 1C-10 mm² IEC01 at each tray joint	—	pcs	Not given	EE-14

7 Inverter room

No.	Item	Specification	Qty	Unit	Basis	Sheet
49	**Steel equipment frame**	Wall-mounted, ≈ 4.5 m W × 1.75 m H, 4 posts; section not given	**1**	set	Drawing	EE-12
50	**AC / DC trunking**	On equipment frame; size not given	**2**	runs	Drawing	EE-12

Points to check in the drawings

No.	Topic	Finding
1	**Sheet list**	GE-01 list does not match the sheets: no perforated-tray layout, no PPR water-pipe layout/detail, no EE-16.
2	**Inverter model**	Written three ways (SUN2000-50KTL-MC0 / -50K-MC0 / -50K-MPO); spec table shows 55 kW rated vs 50 kW max. Confirm model, MPPT and input count.
3	**Inverter AC cable**	EE-01: 4 × 1C-16 mm² + G 10; EE-02/03: 4 × 1C-25 mm² + G 16. 25 mm² is the safe size for a 100 A breaker.
4	**Main feeder**	2 × (4 × 1C-25 mm²) on 250 AT: check ampacity for the installation method; check G 16 mm² against EIT practice for 250 A.
5	**Breaking capacity**	No kA rating given for any MCCB; busbar size not given.
6	**String voltage**	22 × Voc 53.9 V = 1,186 V > 1,000 V inverter limit; only acceptable with optimizers rated for it. Confirm with our supplier.
7	**String labels**	Inverter 2 strings labelled STR-01-xx on EE-03; EE-06 labels differ from the SLD; EE-08 shows 19 modules titled '20 modules/string'.
8	**Cable route**	EE-09 shows IMC conduit on the roof; EE-11/EE-14 show a DC cable tray. No size or length given.
9	**Cleaning water**	PPR pipe appears only in EE-11 sections: no layout, size, tap point or fittings.
10	**Network**	LAN CAT5 on EE-01 vs FTP CAT6 on EE-04.
11	**Surge protection**	SMDB shows a Type 2 50 kA arrester and an SPD behind a 3P fuse (one or two devices?). No DC-side SPD shown.
12	**Earthing**	3 rods Ø 5/8" × 3 m spaced 300 mm (normally ≥ rod length); note on 5 Ω test is ambiguous.
13	**Missing quantities**	No cable lengths, optimizer model, conduit sizes or MDB CT ratio. Mounting quantities are estimated from the EE-05 layout.
14	**North-roof L-feet**	On the north/south faces of the north roof the rails run along the eave, parallel to the roof purlins: each L-foot must land on a purlin. Confirm purlin spacing on site before setting rail positions.

Prepared by Solar ESS Working Group. Quantities marked Est. are to be confirmed on site before ordering.